

Finite Element Formulation for 3D Non-Linear Hyperelastic Material Models at Finite Strain

EMA705 – HW 2

1. Problem Statement

We consider a body occupying the reference domain $\Omega_0 = [0, L_1] \times [0, L_2] \times [0, L_3]$ with $L_1 = 0.10$ m, $L_2 = L_3 = 0.03$ m. The boundary conditions are:

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_D^0 = \{X_1 = 0\}, \quad (1)$$

$$u_1 = \bar{u} = 0.05 \text{ m} \quad \text{on } \Gamma_D^1 = \{X_1 = L_1\}, \quad (2)$$

$$\mathbf{P} \cdot \mathbf{N} = \mathbf{0} \quad \text{on } \Gamma_N = \partial\Omega_0 \setminus (\Gamma_D^0 \cup \Gamma_D^1). \quad (3)$$

2. Kinematics

The deformation map is $\mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X})$. The deformation gradient, its determinant, the right Cauchy–Green tensor, and the left Cauchy–Green tensor are

$$\mathbf{F} = \mathbf{I} + \frac{\partial \mathbf{u}}{\partial \mathbf{X}}, \quad J = \det \mathbf{F}, \quad \mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{B} = \mathbf{F} \mathbf{F}^T. \quad (4)$$

The principal invariants of $\mathbf{C}$ are

$$I_1 = \text{tr}(\mathbf{C}) = F_{iJ} F_{iJ}, \quad I_2 = \frac{1}{2} [I_1^2 - \text{tr}(\mathbf{C}^2)], \quad I_3 = \det(\mathbf{C}) = J^2. \quad (5)$$

The deviatoric (isochoric) invariants are

$$\bar{I}_1 = J^{-2/3} I_1, \quad \bar{I}_2 = J^{-4/3} I_2. \quad (6)$$

3. Constitutive Models

3.1. Model 1: Neo-Hookean

$$W^{\text{NH}} = C_{10}(\bar{I}_1 - 3) + \frac{1}{D_1}(J - 1)^2, \quad C_{10} = 10^{10} \text{ Pa}, \quad D_1 = 10^{-10} \text{ Pa}^{-1}. \quad (7)$$

3.2. Model 2: Mooney–Rivlin

$$W^{\text{MR}} = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{1}{D_1}(J - 1)^2, \quad (8)$$

with $C_{10} = 8 \times 10^9 \text{ Pa}$, $C_{01} = 2 \times 10^9 \text{ Pa}$, $D_1 = 10^{-10} \text{ Pa}^{-1}$.

4. Weak Form (Principle of Virtual Work)

The quasi-static equilibrium in the Total Lagrangian framework requires:

$$\boxed{\mathcal{R}(\mathbf{u}; \mathbf{w}) = \int_{\Omega_0} P_{iJ} \frac{\partial w_i}{\partial X_J} dV = 0 \quad \forall \mathbf{w} \in \mathcal{V}_0.} \quad (9)$$

5. Notation and Useful Derivatives

The following identities are used throughout:

$$\frac{\partial J}{\partial F_{jK}} = J F_{jK}^{-T}, \quad (10)$$

$$\frac{\partial I_1}{\partial F_{jK}} = 2 F_{jK}, \quad (11)$$

$$\frac{\partial I_2}{\partial F_{jK}} = 2(I_1 F_{jK} - (\mathbf{F}\mathbf{C})_{jK}), \quad \text{where } (\mathbf{F}\mathbf{C})_{jK} = F_{jM} C_{MK}, \quad (12)$$

$$\frac{\partial F_{iJ}^{-T}}{\partial F_{jK}} = -F_{jJ}^{-T} F_{iK}^{-T}, \quad (13)$$

$$\frac{\partial (J^{-2/3})}{\partial F_{jK}} = -\frac{2}{3} J^{-2/3} F_{jK}^{-T}, \quad (14)$$

$$\frac{\partial (J^{-4/3})}{\partial F_{jK}} = -\frac{4}{3} J^{-4/3} F_{jK}^{-T}. \quad (15)$$

Additionally, the derivative of the product $(\mathbf{F}\mathbf{C})_{iJ} = F_{iM} F_{nM} F_{nJ}$ is:

$$\frac{\partial (\mathbf{F}\mathbf{C})_{iJ}}{\partial F_{jK}} = \delta_{ij} C_{JK} + F_{iK} F_{jJ} + B_{ij} \delta_{JK}, \quad (16)$$

where $B_{ij} = F_{iM} F_{jM}$ are the components of the left Cauchy–Green tensor $\mathbf{B}$.

6. First Piola–Kirchhoff Stress $\mathbf{P} = \partial W / \partial \mathbf{F}$

The stress is split as $P_{iJ} = P_{iJ}^{\text{iso}} + P_{iJ}^{\text{vol}}$.

6.1. Volumetric contribution (common to both models)

$$P_{iJ}^{\text{vol}} = \frac{2}{D_1} (J - 1) J F_{iJ}^{-T}. \quad (17)$$

6.2. Neo-Hookean isochoric stress

$$\boxed{P_{iJ}^{\text{NH}} = C_{10} J^{-2/3} \left[2 F_{iJ} - \frac{2}{3} I_1 F_{iJ}^{-T} \right] + \frac{2}{D_1} (J - 1) J F_{iJ}^{-T}.} \quad (18)$$

6.3. Mooney–Rivlin stress

$$\boxed{P_{iJ}^{\text{MR}} = P_{iJ}^{\text{NH,iso}} \Big|_{C_{10}} + C_{01} J^{-4/3} \left[2(I_1 F_{iJ} - (\mathbf{F}\mathbf{C})_{iJ}) - \frac{4}{3} I_2 F_{iJ}^{-T} \right] + \frac{2}{D_1} (J - 1) J F_{iJ}^{-T}}. \quad (19)$$

7. FE Discretization

The domain Ω_0 is partitioned into n_{el} elements. Using quadratic (27-node) hexahedral Lagrange shape functions $N_a(\boldsymbol{\xi})$, $a = 1, \dots, 27$, the trial and test fields are

$$u_i^h = \sum_{a=1}^{27} N_a u_{a,i}, \quad w_i^h = \sum_{a=1}^{27} N_a w_{a,i}. \quad (20)$$

The deformation gradient at each quadrature point is

$$F_{iJ} = \delta_{iJ} + \sum_{a=1}^{27} u_{a,i} \frac{\partial N_a}{\partial X_J}, \quad \frac{\partial N_a}{\partial X_J} = \mathcal{J}_{KJ}^{-1} \frac{\partial N_a}{\partial \xi_K}, \quad (21)$$

where $\mathcal{J}_{JK} = \partial X_J / \partial \xi_K$ is the isoparametric Jacobian (not to be confused with $J = \det \mathbf{F}$).

8. Discrete Nonlinear Residual

The element residual vector for node a , component i is:

$$\boxed{R_{ai}^e = \sum_{g=1}^{n_{\text{gp}}} P_{iJ}(\mathbf{F}_g) \frac{\partial N_a}{\partial X_J} \Big|_g \det(\mathcal{J}_g) w_g}, \quad (22)$$

where g indexes the $3 \times 3 \times 3 = 27$ Gauss points and w_g are the quadrature weights. The global system is $\mathbf{R}(\mathbf{U}) = \mathbf{0}$.

9. Tangent Stiffness (Jacobian) for Newton–Raphson

Linearizing the residual about the current state:

$$\mathbf{R}(\mathbf{U}^{(k)} + \Delta \mathbf{U}) \approx \mathbf{R}(\mathbf{U}^{(k)}) + \mathbf{K}_T \Delta \mathbf{U} = \mathbf{0}. \quad (23)$$

The element tangent stiffness is

$$\boxed{K_{ai,bj}^e = \sum_{g=1}^{n_{\text{gp}}} c_{iJjK} \Big|_g \frac{\partial N_a}{\partial X_J} \Big|_g \frac{\partial N_b}{\partial X_K} \Big|_g \det(\mathcal{J}_g) w_g}, \quad (24)$$

where the *first elasticity tensor* (material tangent modulus) is

$$c_{iJjK} = \frac{\partial P_{iJ}}{\partial F_{jK}} = \frac{\partial^2 W}{\partial F_{iJ} \partial F_{jK}}. \quad (25)$$

This tensor possesses the major symmetry $c_{iJjK} = c_{jKiJ}$.

10. Full Analytical Elasticity Tensor

We decompose $\mathcal{C}_{iJjK} = \mathcal{C}_{iJjK}^{\text{iso}} + \mathcal{C}_{iJjK}^{\text{vol}}$.

10.1. Volumetric tangent (common to both models)

Differentiating (17) using (10) and (13):

$$\frac{\partial P_{iJ}^{\text{vol}}}{\partial F_{jK}} = \frac{2}{D_1} \left[\frac{\partial[(J^2 - J)]}{\partial F_{jK}} F_{iJ}^{-T} + (J - 1) J \frac{\partial F_{iJ}^{-T}}{\partial F_{jK}} \right]. \quad (26)$$

Using $\partial(J^2 - J)/\partial F_{jK} = (2J - 1) J F_{jK}^{-T}$ and Eq. (13):

$$\boxed{\mathcal{C}_{iJjK}^{\text{vol}} = \frac{2}{D_1} \left[(2J - 1) J F_{iJ}^{-T} F_{jK}^{-T} - (J - 1) J F_{jJ}^{-T} F_{iK}^{-T} \right]}. \quad (27)$$

10.2. Neo-Hookean isochoric tangent

Define $T_{iJ} = 2 F_{iJ} - \frac{2}{3} I_1 F_{iJ}^{-T}$ so that $P_{iJ}^{\text{NH,iso}} = C_{10} J^{-2/3} T_{iJ}$. By the product rule:

$$\mathcal{C}_{iJjK}^{\text{NH,iso}} = C_{10} \left[-\frac{2}{3} J^{-2/3} F_{jK}^{-T} T_{iJ} + J^{-2/3} \frac{\partial T_{iJ}}{\partial F_{jK}} \right], \quad (28)$$

where

$$\frac{\partial T_{iJ}}{\partial F_{jK}} = 2 \delta_{ij} \delta_{JK} - \frac{4}{3} F_{jK} F_{iJ}^{-T} + \frac{2}{3} I_1 F_{jJ}^{-T} F_{iK}^{-T}. \quad (29)$$

Expanding $F_{jK}^{-T} T_{iJ} = 2 F_{iJ} F_{jK}^{-T} - \frac{2}{3} I_1 F_{iJ}^{-T} F_{jK}^{-T}$ and collecting:

$$\boxed{\mathcal{C}_{iJjK}^{\text{NH,iso}} = C_{10} J^{-2/3} \left[2 \delta_{ij} \delta_{JK} - \frac{4}{3} (F_{iJ} F_{jK}^{-T} + F_{iJ}^{-T} F_{jK}) \right.} \\ \left. + \frac{4}{9} I_1 F_{iJ}^{-T} F_{jK}^{-T} + \frac{2}{3} I_1 F_{jJ}^{-T} F_{iK}^{-T} \right]. \quad (30)$$

The **complete Neo-Hookean tangent** is:

$$\mathcal{C}_{iJjK}^{\text{NH}} = \mathcal{C}_{iJjK}^{\text{NH,iso}} + \mathcal{C}_{iJjK}^{\text{vol}}. \quad (31)$$

10.3. Mooney–Rivlin: additional $\bar{I}_2$ tangent

Define $Q_{iJ} = 2(I_1 F_{iJ} - (\mathbf{FC})_{iJ}) - \frac{4}{3} I_2 F_{iJ}^{-T}$ so that $P_{iJ}^{\bar{I}_2} = C_{01} J^{-4/3} Q_{iJ}$. By the product rule:

$$\mathcal{C}_{iJjK}^{\bar{I}_2} = C_{01} \left[-\frac{4}{3} J^{-4/3} F_{jK}^{-T} Q_{iJ} + J^{-4/3} \frac{\partial Q_{iJ}}{\partial F_{jK}} \right]. \quad (32)$$

The derivative of Q_{iJ} is assembled from (11), (12), (16), and (13):

$$\frac{\partial Q_{iJ}}{\partial F_{jK}} = 2 \left[2 F_{jK} F_{iJ} + I_1 \delta_{ij} \delta_{JK} - \delta_{ij} C_{JK} - F_{iK} F_{jJ} - B_{ij} \delta_{JK} \right] \\ - \frac{8}{3} (I_1 F_{jK} - (\mathbf{FC})_{jK}) F_{iJ}^{-T} + \frac{4}{3} I_2 F_{jJ}^{-T} F_{iK}^{-T}. \quad (33)$$

Expanding $F_{jK}^{-T} Q_{iJ}$:

$$F_{jK}^{-T} Q_{iJ} = 2(I_1 F_{iJ} - (\mathbf{F}\mathbf{C})_{iJ}) F_{jK}^{-T} - \frac{4}{3} I_2 F_{iJ}^{-T} F_{jK}^{-T}, \quad (34)$$

and collecting all terms yields the final result:

$$\begin{aligned} \mathcal{C}_{iJjK}^{\bar{I}_2} = C_{01} J^{-4/3} & \left[4 F_{iJ} F_{jK} + 2 I_1 \delta_{ij} \delta_{JK} - 2 \delta_{ij} C_{JK} - 2 F_{iK} F_{jJ} - 2 B_{ij} \delta_{JK} \right. \\ & - \frac{8}{3} (I_1 F_{iJ} - (\mathbf{F}\mathbf{C})_{iJ}) F_{jK}^{-T} - \frac{8}{3} (I_1 F_{jK} - (\mathbf{F}\mathbf{C})_{jK}) F_{iJ}^{-T} \\ & \left. + \frac{16}{9} I_2 F_{iJ}^{-T} F_{jK}^{-T} + \frac{4}{3} I_2 F_{jJ}^{-T} F_{iK}^{-T} \right]. \end{aligned} \quad (35)$$

The **complete Mooney–Rivlin tangent** is:

$$\mathcal{C}_{iJjK}^{\text{MR}} = \mathcal{C}_{iJjK}^{\text{NH,iso}} \Big|_{C_{10}} + \mathcal{C}_{iJjK}^{\bar{I}_2} + \mathcal{C}_{iJjK}^{\text{vol}}. \quad (36)$$

11. Newton–Raphson Solution Algorithm

The prescribed displacement $\bar{u}$ is applied in N_{step} increments $\Delta \bar{u} = \bar{u}/N_{\text{step}}$. At each load step n :

1. Set the predictor $\mathbf{U}^{(0)} = \mathbf{U}_{n-1}$. Apply the Dirichlet increment on the constrained DOFs.
2. **Newton iteration** ($k = 0, 1, 2, \dots$):
 - (a) Compute F_{iJ} at all Gauss points from $\mathbf{U}^{(k)}$.
 - (b) Evaluate $P_{iJ}(\mathbf{F})$ and $\mathcal{C}_{iJjK}(\mathbf{F})$ using the boxed expressions above.
 - (c) Assemble $\mathbf{R}^{(k)}$ from (22) and $\mathbf{K}_T^{(k)}$ from (24).
 - (d) Apply Dirichlet BCs (zero rows/columns of $\mathbf{K}_T$; zero corresponding $\mathbf{R}$ entries).
 - (e) Solve $\mathbf{K}_T^{(k)} \Delta \mathbf{U} = -\mathbf{R}^{(k)}$.
 - (f) Update $\mathbf{U}^{(k+1)} = \mathbf{U}^{(k)} + \Delta \mathbf{U}$.
 - (g) Check convergence: $\|\mathbf{R}^{(k)}\|/\|\mathbf{R}^{(0)}\| < \text{tol}$ (e.g., 10^{-8}).
3. Set $\mathbf{U}_n = \mathbf{U}^{(k+1)}$. Proceed to next load step.